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Domain: Security Operations Center (SOC)
Topic: SOC Alert Management and Incident Response Report
Description:
This report covers the fundamental concepts of Security Operations Center (SOC) operations, including alert priority levels, incident classification, and the basic incident response lifecycle. It explains how SOC analysts prioritize alerts using factors such as severity, asset criticality, exploit likelihood, and business impact, along with the use of CVSS scoring.
The report also includes practical SOC activities such as alert classification, alert triage, incident ticket creation, threat intelligence validation, evidence preservation, and a simulated alert-to-response workflow to demonstrate real SOC operational practices.

Theoretical Report

1. Alert Priority Levels

Introduction
In a Security Operations Center (SOC), a large number of security alerts are generated daily by monitoring tools such as SIEM, IDS/IPS, and endpoint security systems. Since not all alerts represent serious threats, SOC analysts must prioritize them based on severity and potential impact. Alert priority levels help security teams quickly identify critical threats and respond effectively to protect organizational assets.

Priority Levels
Security alerts are commonly categorized into four priority levels:
Critical:
These alerts represent the most severe threats and require immediate action. They usually indicate active attacks or system compromise.
Example: Ransomware attack or exploitation of a critical vulnerability.
High:
High priority alerts indicate serious suspicious activity that could lead to system compromise if not addressed quickly.
Example: Unauthorized administrative access.
Medium:
Medium priority alerts represent potentially suspicious activities that require investigation but may not pose immediate risk.
Example: Multiple failed login attempts.

Low:
Low priority alerts represent informational or minor events with minimal impact.
Example: Network port scanning.

Alert Prioritization Criteria
SOC analysts determine alert priority using several factors:

Asset Criticality:
Alerts affecting critical systems such as production servers are given higher priority than those affecting test systems.

Exploit Likelihood:
Vulnerabilities with publicly available exploits are more likely to be targeted by attackers.

Business Impact:
If a successful attack could cause financial loss, service disruption, or data exposure, the alert receives higher priority.

CVSS Scoring System
The **Common Vulnerability Scoring System (CVSS)** is used to measure the severity of vulnerabilities. CVSS assigns a score between **0 and 10** based on risk level.

CVSS Score	Severity
0 – 3.9	Low
4 – 6.9	Medium
7 – 8.9	High
9 – 10	Critical

Example: Log4Shell Vulnerability

The **Log4Shell vulnerability (CVE-2021-44228)** in Apache Log4j allowed attackers to execute remote code on vulnerable systems. It received a **CVSS score of 9.8**, which classified it as a **Critical severity vulnerability**, requiring immediate patching and mitigation.

2. Incident Classification

Introduction

Incident classification is the process of categorizing security incidents based on their type, impact, and attack method. In a Security Operations Center (SOC), proper classification helps analysts quickly understand the nature of an incident and apply the correct response procedures. Standardized classification also improves communication, reporting, and investigation efficiency.

Types of Security Incidents

Security incidents can be classified into several common categories:

Malware:
Malicious software such as viruses, ransomware, or trojans that compromise systems or data.

Phishing:
Fraudulent emails or messages designed to trick users into revealing sensitive information or clicking malicious links.

DDoS (Distributed Denial of Service):
An attack that floods a server or network with traffic to disrupt services.

Insider Threat:
Security risks caused by employees or trusted users who misuse their access.

Data Exfiltration:
Unauthorized transfer of sensitive data outside the organization.

Incident Classification Frameworks

SOC teams often use standard frameworks to classify and analyze incidents.

MITRE ATT&CK:
A widely used framework that maps attacker tactics and techniques.
Example: **T1566 – Phishing**

ENISA Incident Taxonomy:
Provides standardized categories for different types of cybersecurity incidents.

VERIS Framework:
A structured method used to record and share information about security incidents.

Contextual Metadata

To improve incident analysis, SOC analysts add contextual information such as:

- Affected system or asset
- Timestamp of the event
- Source and destination IP addresses
- Indicators of Compromise (IOCs) such as malicious file hashes or URLs

This additional data helps analysts understand the attack and perform effective investigations.

3. Basic Incident Response

Introduction

Incident response is the process of detecting, analyzing, and responding to cybersecurity incidents in order to minimize damage and restore normal operations. In a Security Operations Center (SOC), incident response helps security teams handle threats effectively and protect organizational systems and data. A structured incident response process ensures that security events are managed quickly and efficiently.

Incident Response Lifecycle

The incident response process generally follows a structured lifecycle based on industry standards such as the **NIST Incident Handling Guide (SP 800-61)**.

Preparation:
This phase focuses on preparing the organization to handle incidents. It includes creating security policies, response plans, and incident response playbooks.

Identification:
During this phase, SOC analysts detect and analyze potential security incidents using monitoring tools, logs, and alerts.

Containment:
The goal of this phase is to limit the spread of the attack. This may include isolating affected systems or blocking malicious IP addresses.

Eradication:
In this phase, the root cause of the incident is removed, such as deleting malware or patching vulnerabilities.

Recovery:
Systems are restored to normal operation after ensuring that the threat has been completely removed.

Lessons Learned:
After resolving the incident, teams review the event to improve security processes and prevent similar incidents in the future.

Evidence Preservation

During incident response, it is important to preserve digital evidence for investigation. SOC analysts collect system logs, memory dumps, and file hashes to analyze the attack while maintaining the integrity of the evidence.

Practical Report

1. Alert Classification

- In this practical task, an alert classification table was created using Google Sheets to categorize different types of security alerts based on their priority and attack technique. Each alert was mapped with the corresponding MITRE ATT&CK technique to understand the nature of the attack.

  For example, phishing emails were mapped with MITRE technique T1566, brute force SSH attempts with T1110, Log4Shell exploitation with T1190, and port scanning with T1046. This classification helps SOC analysts quickly identify the type of threat and prioritize alerts for further investigation.

  The alert classification table demonstrates how SOC teams organize and analyze alerts generated by security monitoring tools.

	A	B	C	D	E
1	**Alert ID**	**Type**	**Priority**	**MITRE Technique**	
2	1	Phishing Email	High	T1566	
3	2	Brute Force SSH	Medium	T1110	
4	3	Log4Shell Exploit	Critical	T1190	
5	4	Port Scan	Low	T1046	
6					

Figure 1: Alert Classification Table mapped with MITRE ATT&CK techniques

2. Alert Prioritization

- **Alert Prioritization Using CVSS** :-

  In this task, security alerts were prioritized based on their severity using the Common Vulnerability Scoring System (CVSS). CVSS helps security analysts measure the risk level of vulnerabilities and security incidents.

  Alerts with higher CVSS scores are treated as more critical and require immediate attention. For example, the Log4Shell vulnerability has a CVSS score of 9.8, which classifies it as a critical threat. Medium and low severity alerts such as brute force login attempts and port scanning activities are investigated accordingly based on their risk level.

  This prioritization process helps SOC analysts focus on the most dangerous threats first and improves incident response efficiency.

	A	B	C	D	E
1	**Alert**	**Description**	**CVSS Score**	**Priority**	
2	Log4Shell	Remote Code Execution	9.8	Critical	
3	Brute Force Login	Multiple login attempts	6.5	Medium	
4	Port Scan	Network scanning activity	3.5	Low	
5					
6					

Figure 2: Alert Prioritization Table based on CVSS Scores

3. Incident Ticket Creation

Incident Ticket

Title: [Critical] Ransomware Detected on Server-X

Description:
Suspicious executable file "crypto_locker.exe" detected on the system.
Unusual network communication observed from IP address 192.168.1.50.

Priority: Critical

Assignee: SOC Analyst

Status: Open

- **Incident Ticket Documentation :-** In this step, a sample incident ticket was created to document a critical security alert. Incident tickets are used in Security Operations Centers (SOC) to track and manage security incidents during investigation.

  The ticket contains important details such as the incident title, description, priority level, assigned analyst, and current status. In this example, a ransomware activity was detected on a server and documented as a critical incident for further investigation.

  Incident documentation helps SOC teams maintain proper records of security events and ensures effective coordination during incident response.

4. Escalation Email

Dear Tier 2 SOC Team,

A critical security alert has been detected on Server-X indicating possible ransomware activity. Initial investigation identified a suspicious executable file named crypto_locker.exe and unusual network communication from IP address 192.168.1.50. The affected system has been isolated to prevent further spread. Indicators of compromise have been collected and require further analysis.

Please review the incident and proceed with further investigation.

Regards
SOC Analyst

➢ **Incident Escalation Process**

In this step, a sample escalation email was created to notify the Tier-2 SOC team about a critical security incident. Escalation is an important part of SOC operations where serious alerts are forwarded to senior analysts for deeper investigation.

The email summarizes the detected threat, indicators of compromise, and the actions taken by the analyst. This process ensures that critical incidents receive immediate attention and are handled efficiently.

5. Incident Response Template

Incident Response Report

1. Executive Summary
A suspicious phishing email was detected targeting multiple users in the organization. The SOC team analyzed the email and confirmed it contained a malicious link.

2. Timeline
14:00 – Phishing email detected by security monitoring system
14:10 – SOC analyst started investigation
14:20 – Malicious link confirmed
14:30 – Email blocked and users notified

3. Impact Analysis
The phishing email attempted to steal user credentials. No users clicked the link and no systems were compromised.

4. Remediation Steps
The malicious domain was blocked in the firewall and users were advised to avoid similar emails.

5. Lessons Learned
User awareness training and improved email filtering rules can help prevent similar attacks.

- **Incident Response Documentation**
  In this task, an incident response template was created to document a simulated phishing incident. Proper documentation helps SOC teams record investigation steps, timeline of events, impact analysis, and remediation actions.

  Structured incident reports improve communication between security teams and help organizations learn from security incidents to strengthen their defenses.

6. Investigation Timeline

	A	B
1	**Timestamp**	**Action**
2	3/6/2026 16:20	Suspicious activity detected by monitoring system
3	3/6/2026 16:25	SOC analyst started investigation
4	3/6/2026 16:30	Malicious file identified
5	3/6/2026 16:35	Affected system isolated to prevent further spread
6		
7		

Figure 3: Incident Investigation Timeline

- **Incident Investigation Timeline:**
  During the investigation process, a timeline of events was created to record the actions taken by the SOC team. Maintaining a timeline helps analysts track the sequence of activities and understand how the incident progressed.

  The timeline includes important events such as the detection of suspicious activity, investigation steps, identification of malicious files, and containment actions taken to isolate the affected system.

7. Phishing Investigation Checklist

☐ Verify email sender address
☐ Analyze email headers
☐ Check suspicious links using VirusTotal
☐ Identify affected users
☐ Block malicious domain or IP
☐ Notify users about phishing attempt

➢ **Phishing Investigation Checklist**
During phishing incident investigations, SOC analysts follow a structured checklist to ensure that all necessary steps are completed. The checklist includes verifying the email sender, analyzing email headers, checking suspicious links using threat intelligence platforms, identifying affected users, and blocking malicious domains.

Using a checklist helps security teams perform consistent investigations and reduces the risk of missing important analysis steps.

8. Post-Mortem Summary

The simulated phishing incident highlighted the importance of early detection and quick response. The SOC team successfully identified the malicious email and prevented users from accessing the harmful link.

To prevent similar incidents in the future, stronger email filtering rules and regular security awareness training for employees are recommended.

9. Alert Triage Table

	A	B	C	D	E
1	Alert ID	Description	Source IP	Priority	Status
2	2	Brute-force SSH attempts	192.168.1.100	Medium	Open
3					

Figure 4: Alert Triage Analysis Table

- **Alert Triage Analysis**
  Alert triage is the process of analyzing security alerts to determine their severity and validity. In this step, a sample alert related to brute-force SSH login attempts was analyzed.

  The alert included details such as the source IP address, description of the activity, priority level, and current status. Triage helps SOC analysts quickly identify potential threats and decide whether further investigation or escalation is required.

10. Threat Intelligence Validation

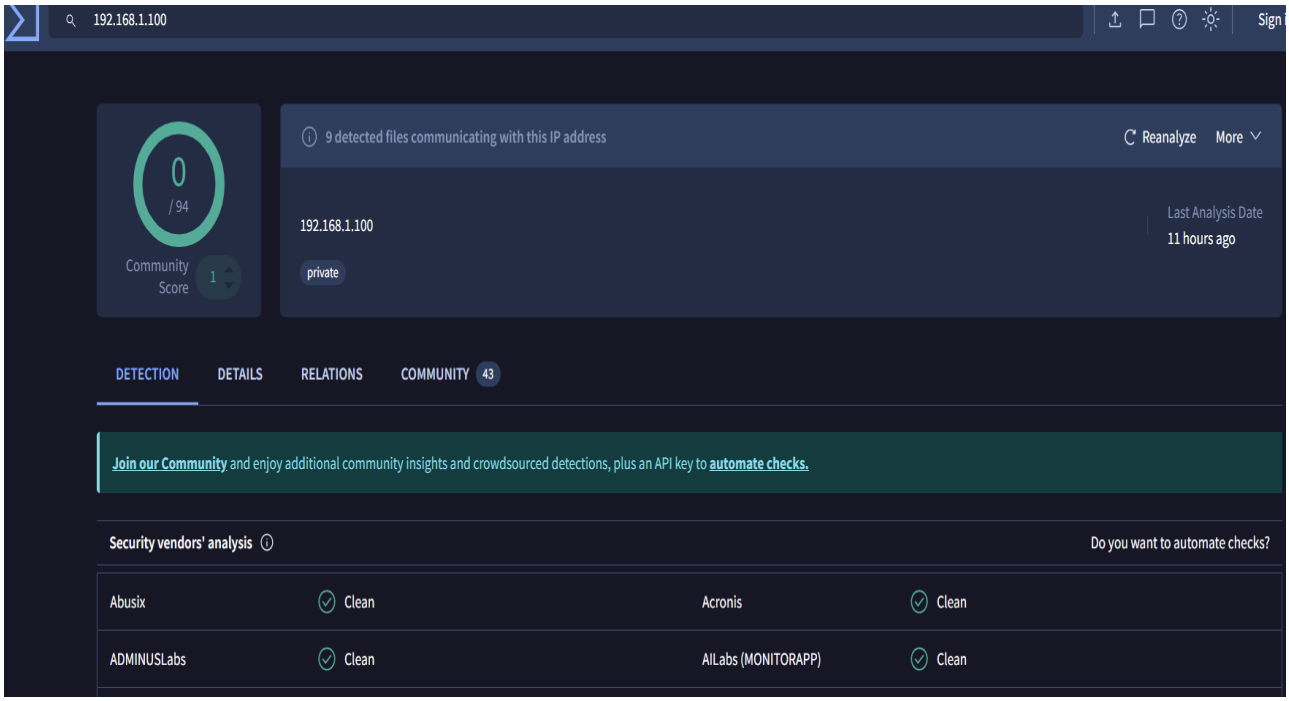

Figure 5: Threat Intelligence Check using VirusTotal

- **Threat Intelligence Validation**
  Threat intelligence platforms such as VirusTotal and AlienVault OTX are commonly used by SOC analysts to verify indicators of compromise (IOCs). In this task, the suspicious IP address 192.168.1.100 was checked using a threat intelligence platform.

  The analysis helps determine whether the IP address has been previously associated with malicious activities such as brute-force attacks or malware distribution. Threat intelligence validation assists SOC analysts in confirming potential threats and improving investigation accuracy.

11. Network Evidence Collection

```
┌──(kali㉿kali)-[~]
└─$ ss -tulnp
Netid  State   Recv-Q  Send-Q   Local Address:Port   Peer Address:Port   Process
tcp    LISTEN  0       5            0.0.0.0:8080         0.0.0.0:*       users:(("python3",pid=4001,fd=3))
```

Figure 6: Network Connections Evidence collected using ss command

- **Network Evidence Collection**

  During incident investigation, SOC analysts collect network evidence to identify suspicious connections and active services on a system. In this task, the Linux command "ss -tulnp" was used to display active network ports and connections.

  This information helps analysts detect unusual network activity and identify potential threats communicating with the system.

12. Digital Evidence Hashing

```
┌──(kali㉿kali)-[~]
└─$ echo "SOC Investigation Sample File" > evidence.txt

┌──(kali㉿kali)-[~]
└─$ sha256sum evidence.txt
f7f192aa652afc58462530408f4371e9bde34cda21bd52d51748a428bb5f99ea  evidence.txt
```

Figure 7: SHA-256 Hash generated for digital evidence

- **Digital Evidence Hashing**

  During the investigation process, digital evidence hashing was performed to ensure the integrity of collected files. A sample evidence file was created and the SHA-256 hashing algorithm was used to generate its hash value.

  Hashing helps SOC analysts verify that the evidence has not been modified during the investigation process. SHA-256 is widely used in digital forensics to maintain the integrity and authenticity of evidence.

13. Attack Simulation Scenario

In this simulated scenario, a vulnerability exploitation attempt was performed against a vulnerable server. The attacker attempted to exploit a known vulnerability in the VSFTPD service.

The security monitoring system detected suspicious activity originating from IP address 192.168.1.100. The alert indicated a possible exploitation attempt targeting the FTP service.

The SOC analyst investigated the alert and confirmed that it was related to a potential remote exploitation attempt.

14. Detection and Triage

	A	B	C	D
1	**Timestamp**	**Source IP**	**Alert Description**	**MITRE Technique**
2	3/6/2026 17:10	192.168.1.100	VSFTPD exploitation attempt detected	T1190
3				

Figure 8: Detection and Alert Analysis Table

- **Attack Detection and Alert Analysis**

During the attack simulation, the monitoring system generated an alert indicating a possible exploitation attempt targeting the FTP service. The alert was analyzed by the SOC analyst to identify the source IP address and the nature of the attack.

The alert details were documented including timestamp, source IP, and the corresponding MITRE ATT&CK technique. This information helps SOC teams understand the attack pattern and respond effectively.

15. Incident Response Action

After detecting the suspicious exploitation attempt, the SOC analyst immediately initiated containment procedures. The affected system was isolated from the network to prevent further compromise.

The attacker's IP address (192.168.1.100) was blocked through firewall rules to stop malicious communication. Additional monitoring was performed to ensure that no further suspicious activity was observed on the network.

16. Final Incident Report

Executive Summary
A simulated exploitation attempt targeting an FTP service was detected by the monitoring system. The attack originated from IP address 192.168.1.100 and was identified as a possible remote exploitation attempt.

Timeline
17:05 – Suspicious activity detected by monitoring system
17:10 – SOC analyst started investigation
17:15 – Exploitation attempt confirmed
17:20 – Affected system isolated and attacker IP blocked

Recommendations
To prevent similar attacks in the future, it is recommended to regularly update vulnerable services, implement stronger network monitoring rules, and conduct regular security assessments.

❖ Conclusion

This report presented both theoretical knowledge and practical implementation of Security Operations Center (SOC) activities. The theoretical section explained important SOC concepts such as alert priority levels, incident classification, and the incident response lifecycle.

In the practical section, different SOC tasks were performed including alert classification, alert prioritization, incident ticket creation, threat intelligence validation, alert triage, evidence preservation, and a simulated attack-to-response workflow.

These activities demonstrate how SOC analysts monitor security events, investigate alerts, and respond to potential threats in a structured manner. The practical exercises helped in understanding real SOC operations and the importance of proper documentation, analysis, and incident response procedures for maintaining organizational security.